FOSS4G, Brussels, October 2026

From Orthophotos to Objects: Building a GeoAI Change-Detection Pipeline with Open-Source Tools at NGI

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B_ROBOTS

NGI Nationaal Geografisch Instituut

IGN Institut Géographique National

GeoAI POC | INTRODUCTION

What is GeoAI?

GeoAI combines Artificial Intelligence with geospatial data to automatically extract, analyse and understand information about our changing world.



CORE AI TASKS IN GEOSPATIAL ANALYSIS

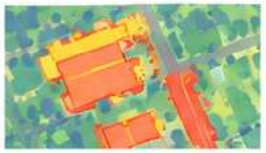
Image recognition
Classify imagery



Object detection
Locate objects



Semantic segmentation
Classify pixels



Instance segmentation
Identify individual objects



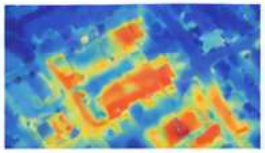
Image translation
Transform imagery



Change detection
Detect changes over time



Pixel regression
Predict continuous values

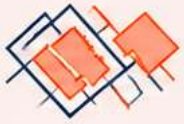


THE EVOLUTION OF GEOAI

1. **TRADITIONAL METHODS**
   Rules, image processing and manual feature engineering.
2. **MACHINE LEARNING**
   Classification using engineered features.
3. **DEEP LEARNING**
   CNNs, segmentation and automated feature extraction.
4. **FOUNDATION MODELS**
   Pretrained models, transfer learning and adaptable GeoAI.

OUR SCOPE IN THIS POC

1. **Feature extraction**
   Automatically identify and extract geographic objects from imagery.



2. **Change detection**
   Identify changes between imagery dates to support cartographic updates.

NGI IGN

GeoAI POC

GeoAI through collaboration

NGI works together with Kadaster (NL), EuroSDR and B_Robots to exchange knowledge and develop practical GeoAI applications for cartography.



Kadaster (Netherlands)

KNOWLEDGE EXCHANGE IN CARTOGRAPHY

- Exchange of experience on GeoAI use cases in cartography
- Sharing data, methods and lessons learned
- Mutual inspiration on implementation and operational use

EuroSDR

EUROPEAN KNOWLEDGE SHARING

- Participation in the GeoAI SIG and related workshops
- Exchange on models, data and methods across European mapping agencies
- Contributing to a European model catalogue and LLMs in action



B_Robots

TECHNOLOGY PARTNER FOR THIS POC

- Development of the change detection pipeline
- Expertise in AI, data engineering and geospatial workflows
- From experimentation to a working prototype

KEY TAKEAWAY

Collaboration with Kadaster, **EuroSDR** and **B_Robots** accelerates knowledge sharing and the development of practical GeoAI solutions for national mapping agencies.

GeoAI POC

The journey so far

From a broad question to a working change-detection pipeline

We started with a broad question: **can AI help cartographers?**
Over the past months, we moved from understanding the process to building and training an actual change-detection pipeline. Along the way, the main challenge shifted from technology and infrastructure to data quality.



DISCOVER	DEFINE	ENABLE	BUILD	LEARN	IMPROVE
Understand the process, users and opportunity	Select the use case and technical approach	Set up IT/Azure environment and secure data access	Develop the change-detection pipeline and train the model	Validate performance and uncover data issues	Use model output to improve training data and retrain
Interviews with cartographers, QC and data experts	Change detection first, classification second	Infrastructure and data access took longer than planned	Automated pipeline and first model training completed	Mismatch between database and imagery years affects training data quality	Re-labelling based on model output improves detection performance

June 2026 — Today

GeoAI POC

We started with the cartographer, not the model

Understanding where AI can add value in the cartographic workflow

Before selecting a technical approach, we first explored the cartographic workflow to understand where AI could support cartographers and reduce manual effort.



UNDERSTAND THE PROCESS

Interviews with cartographers and QC to understand how maps are updated and where manual effort is concentrated.

June 2026

EXPLORE THE DATA

Understand available aerial imagery, ITGI data, existing models and the differences between regions and acquisition years.

IDENTIFY OPPORTUNITIES

Explore where AI could support cartographers by reducing repetitive work and directing attention to relevant changes.

DEFINE THE APPROACH

Select the use case and technical approach.
Change detection first, classification second.

BUILD AND VALIDATE

Set up the environment and develop the initial change-detection pipeline to test and validate performance.

Today

KEY TAKEAWAY

The goal is not to replace cartographic judgement, but to help cartographers focus their attention where it matters.

GeoAI POC

Change detection emerged as the first use case

From the opportunities identified during discovery, change detection was selected as the first POC to explore.



WHY CHANGE DETECTION?

- Focuses attention on areas that actually changed.
- Can work across different object types (e.g. buildings, roads).
- Reduces the need to scan unchanged areas.
- Keeps the cartographer in the decision loop.

KEY PRINCIPLE

Detect change first. Classify second.

GeoAI POC

Before training a model, we first had to enable it

Setting up the environment and getting access to the required NGI data took longer than anticipated.



EXPECTED

Data → Build → Train → Test

WHAT WE ENCOUNTERED

Azure setup → Access & permissions → Data access → Data transfer → Real data

MORE TIME THAN ANTICIPATED

Setting up the Azure environment, securing access and transferring the required NGI data took longer than planned.

PROGRESS CONTINUED

Development continued on limited data while infrastructure and access issues were being resolved.

OUTCOME

By the end of this phase, the POC environment was operational and the focus could shift from infrastructure to the data itself.

KEY TAKEAWAY

Enabling the right data and infrastructure is a critical step to build reliable AI models.

GeoAI POC

From experiments to a working pipeline

Using aerial imagery and reference data, we built an automated change-detection pipeline and trained the first model on NGI data.



2018 imagery



2024 imagery



Prepare & align data



Harmonize imagery and database versions and prepare training data.

Detect differences



Identify changed areas between image dates.

Train / run model

Train the model on NGI data and run inference at scale.

Change candidates



List of potential changes (buildings, other objects).

Cartographer review



Cartographers review and decide what to update in the database.

KEY ACHIEVEMENTS IN THIS PHASE

- Automated data preparation and change-detection pipeline.
- First model training completed on NGI data.
- Initial results available for evaluation.

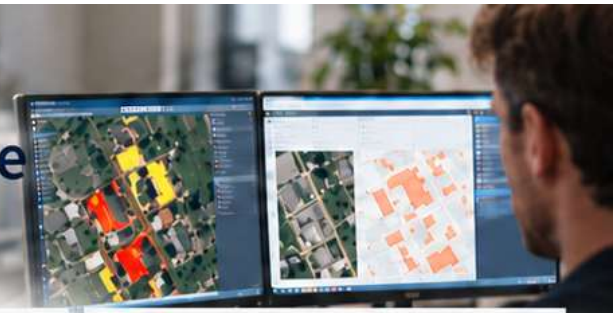
KEY TAKEAWAY

We now have a working change-detection pipeline and first model results to evaluate.

GeoAI POC

The technology behind the change-detection pipeline

An **open-source** geospatial pipeline combining geospatial data, automated processing, deep learning and cloud infrastructure to detect and analyse changes at scale.



1 GEOSPATIAL DATA (OPEN SOURCE)

Aerial imagery + ITGI / reference data

Input from different years is prepared, aligned and transformed into training and inference data.



2 CHANGE DETECTION MODEL (OPEN SOURCE)

Deep learning for bi-temporal imagery

The model compares imagery from two points in time and predicts where relevant changes occurred.

FOTBCD

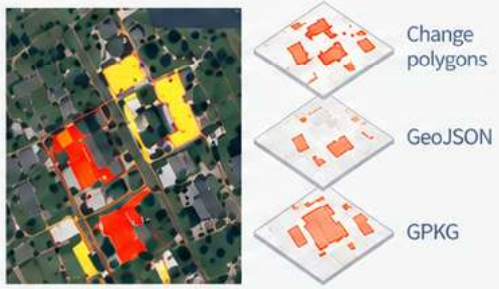
Pretrained model
+ fine-tuned on Belgian / NGI data



3 GEO PROCESSING & PIPELINE (OPEN SOURCE)

Python-based automated processing

Geospatial processing turns raw imagery and database information into model-ready data and converts predictions into usable geographic outputs.



4 AZURE INFRASTRUCTURE

Compute, storage & model training

Azure provides the environment and GPU capacity required to process the data, train models and run experiments.

Compute (GPU) | Storage (data & models) | Model training & experimentation

TECHNOLOGY STACK (OPEN SOURCE)

Key open-source tools used in the POC.

Python | PyTorch | FOTBCD | GeoPandas / GDAL | GeoJSON / GPKG | Azure VM / GPU

KEY TAKEAWAY

The POC is not a single AI model — it is an end-to-end **open-source** GeoAI pipeline combining geospatial data, automated processing, deep learning and cloud infrastructure.

GeoAI POC

The first results uncovered a data challenge

Aerial imagery and database versions do not always represent exactly the same point in time.



2018
Aerial imagery





2018 database

2024
Aerial imagery





2024 database

IMPACT ON MODEL PERFORMANCE

Using correctly aligned imagery dates and database versions leads to significantly better change-detection results.

Metric	Before correction (v1)	After correction (v2)
IoU	0.3105	**0.4395**
F1	0.4738	**0.6107**
mIoU	0.6480	**0.7136**
Precision	0.6936	**0.7109**
Recall	0.3598	**0.5352**

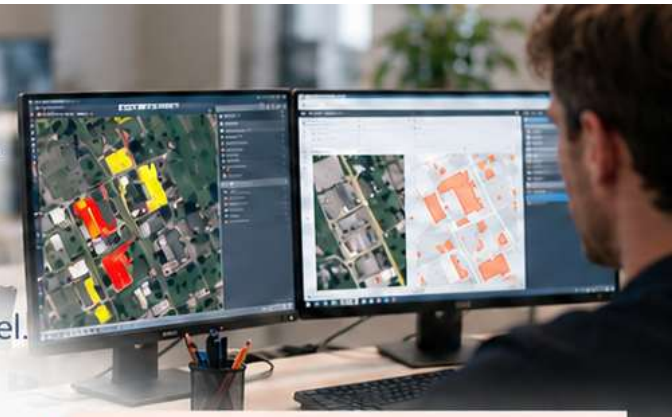
KEY TAKEAWAY

Consistent alignment between imagery dates and database versions is essential for reliable training and evaluation.

GeoAI POC

Today: using the model to improve the data — and the data to improve the model

The current model helps us identify inconsistencies in the reference data.
After cartographic review and correction, the improved data is used to retrain the model.



1 CURRENT MODEL

Run the current change-detection model on new data.



2 IDENTIFY INCONSISTENCIES

Use model predictions to locate potential mismatches in the reference data.



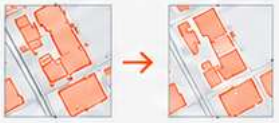
3 CARTOGRAPHIC REVIEW

Cartographers check flagged cases against aerial imagery and mapping specifications.



4 CORRECT THE DATA

Resolve confirmed inconsistencies in the source data.



5 REGENERATE LABELS

Create improved training data from the corrected sources.



6 RETRAIN THE MODEL

Train the next model iteration on the improved reference data.



IMPROVING DATA & MODEL TOGETHER

WHERE WE ARE TODAY

We are currently using model output to detect and resolve data inconsistencies. The next retraining cycle will use this improved reference data.

KEY ACTIVITIES NOW

- Run the current model and analyze predictions
- Identify and prioritize data inconsistencies
- Cartographic review and correction
- Regenerate training labels
- Retrain the model and evaluate results

KEY TAKEAWAY

The model and the data now improve each other in an iterative feedback loop.

GeoAI POC

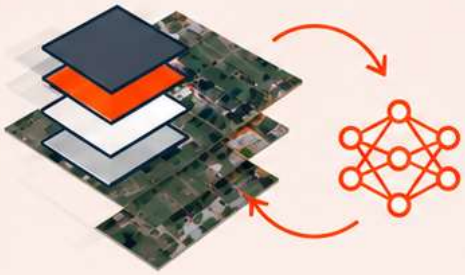
From POC to operational use

The next phase is about turning the improved model into a solution cartographers can use in their daily workflow.



1 RETRAIN THE MODEL (OPEN SOURCE)

Train the next model iteration using the improved and corrected reference data.



2 PRODUCTIONIZE (OPEN SOURCE)

Deploy the model into an operational environment and integrate it into a usable workflow.



3 TEST WITH CARTOGRAPHERS

Test on real cases, validate the results and collect feedback from cartographers.



4 ADOPT IN DAILY WORK

Integrate the solution into the cartographic workflow and progressively move towards daily use.



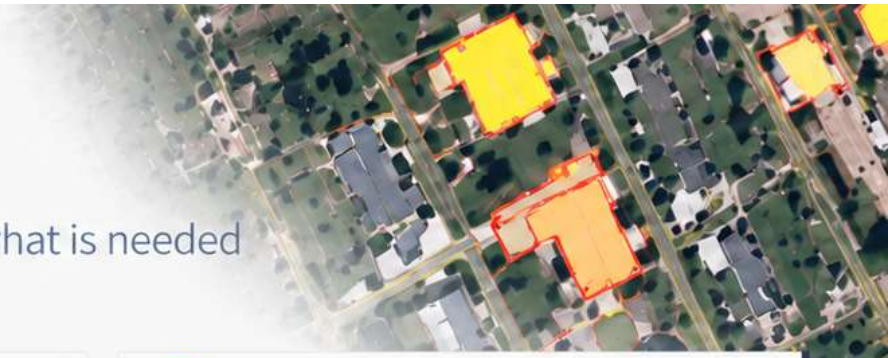
KEY TAKEAWAY

The next challenge is turning the improved model into a solution that creates value in the daily work of cartographers.

GeoAI POC

Key learnings and next steps

The POC has delivered more than a model — it has shown what is needed to make GeoAI work in practice.



1 AI CAN SUPPORT THE CARTOGRAPHER

Change detection can help focus attention on areas that are likely to require review. The objective remains to support cartographic judgement, not replace it.



2 GOOD AI STARTS WITH GOOD DATA

The experiments revealed inconsistencies between imagery dates and database states. Improving this reference data is now an integral part of improving the model.



3 A MODEL IS ONLY THE BEGINNING

Real value will come when the model is operationalized, tested with cartographers and integrated into their daily workflow.



KEY TAKEAWAY

From asking “Can AI help cartographers?” to building the foundations for an AI-assisted cartographic workflow.

GeoAI POC

Q&A

Any questions?

